

INFERENCE STATISTICS: ESTIMATION (CONFIDENCE INTERVAL)

Inferential statistics is a process that infers results from a population using sampling. This process is conventionally divided into two parts: **Estimation** and **Hypothesis**

In the **estimation** process, we infer numerical results about population parameters. As to **hypothesis testing**, we are interested in the problem of choosing one hypothesis from two opposite hypotheses related to the population parameters or population distribution.

A sample statistic that is used for the estimation of an unknown population parameter is called an **estimator**, its value is called an **estimated value**.

Population Parameter (θ)	Estimator $\hat{\theta}$	Estimate value $\hat{\theta}$
Mean (μ)	$\bar{X}$	$\bar{x}$
Variance (σ^2)	S^2	s^2
Standard Deviation (σ)	S	s
Proportion (p)	$\hat{P}$	$\hat{p}$
Difference of Two Means ($\mu_1 - \mu_2$)	$\bar{X}_1 - \bar{X}_2$	$\bar{x}_1 - \bar{x}_2$

A. THE PROPERTIES OF ESTIMATORS

B.

1) Unbiased Estimators: A statistic $\hat{\theta}$ is an **unbiased estimator** of the population parameter θ if and only if $E(\hat{\theta}) = \theta$.

Theorem: If X has a binomial distribution with the parameters n and p , the sample proportion $\hat{P} = X/n$ is an unbiased estimator of the population proportion.

Proof:

$$E(X) = np$$

$$E(\hat{P}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}np = p$$

Theorem: If S^2 is the variance of a random sample selected from an infinite population with the finite variance σ^2 , then

$$E(S^2) = \sigma^2.$$

Proof:

$$\begin{aligned}
E(S^2) &= E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X} + \mu - \mu)^2 \right] \\
&= E \left[\frac{1}{n-1} \sum_{i=1}^n [(X_i - \mu) - (\bar{X} - \mu)]^2 \right] \\
&= E \left[\frac{1}{n-1} \sum_{i=1}^n [(X_i - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu) + (\bar{X} - \mu)^2] \right] \\
&= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu) + n(\bar{X} - \mu)^2 \right) \right] \\
&= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \left(\frac{n}{n} \sum_{i=1}^n X_i - n\mu \right) + n(\bar{X} - \mu)^2 \right) \right] \\
&= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu)(n\bar{X} - n\mu) + n(\bar{X} - \mu)^2 \right) \right] \\
&= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right) \right] = \frac{1}{n-1} \left(\sum_{i=1}^n E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2] \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n \sigma^2 - n\sigma_{\bar{X}}^2 \right) = \frac{1}{n-1} \left(\sum_{i=1}^n \sigma^2 - n \frac{\sigma^2}{n} \right) = \frac{1}{n-1} (n\sigma^2 - \sigma^2) = \sigma^2
\end{aligned}$$

As shown in the proof, the sample variance S^2 is an unbiased estimator when the denominator of its formula is defined by $n-1$ instead of n .

2) Efficiency: If $E(\hat{\theta}_1) = \theta$, $E(\hat{\theta}_2) = \theta$ and $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$, then it is said that **$\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$** .

The relative efficiency of $\hat{\theta}_1$ with respect to $\hat{\theta}_2$ is:

$$\text{Relative Efficiency} = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

If

$$E(\hat{\theta}) = \theta \text{ and } \text{Var}(\hat{\theta}) = \frac{1}{nE \left[\left(\frac{\partial \ln f(x)}{\partial \theta} \right)^2 \right]}$$

then $\hat{\theta}$ is said to be **unbiased estimator with minimum variance** of θ or **the most efficient estimator** of θ .

Example: Show that $\bar{X}$ is the unbiased estimator with minimum variance of μ which is the mean of a normal distributed population.

Since the normal distribution is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty,$$

it follows that

$$\ln f(x) = -\ln \sigma\sqrt{2\pi} - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 \quad \frac{\partial \ln f(x)}{\partial \mu} = \frac{1}{\sigma}\left(\frac{x-\mu}{\sigma}\right)$$

$$E\left[\left(\frac{\partial \ln f(x)}{\partial \mu}\right)^2\right] = \frac{1}{\sigma^2} E\left[\left(\frac{x-\mu}{\sigma}\right)^2\right] = \frac{1}{\sigma^2}$$

$$\text{Var}(\bar{X}) = \frac{1}{nE\left[\left(\frac{\partial \ln f(x)}{\partial \theta}\right)^2\right]} = \frac{\sigma^2}{n}$$

The estimation process is divided into two stages: **Point Estimation** and **Interval Estimation**

B. POINT ESTIMATION

If the estimation of a population is given as only one numerical value, this process is called **point estimation**.

Example: A computer firm produces graphic cards. 5 Graphic cards are randomly selected. If the lasting times of these cards are 8, 11, 5, 7 and 19 months, compute point estimations for the mean and variance of lasting times of all produced graphic cards in the firm.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{50}{5} = 10$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = 30$$

C. INTERVAL ESTIMATION

The probability of occurring a single point for the continuous distributions such as normal, t or χ^2 (Chi-square) is theoretically zero. More clearly, if θ is a population parameter, $\hat{\theta}$ is an estimator and $\hat{\theta}$ is the estimate value (point estimation) of the population parameter, then $P(\theta=\hat{\theta})=0$. Thus, the use of a point estimation is not confidential (zero confidence). To increase confidence level, we can form an error interval about $\hat{\theta}$ by assuming that the estimation includes an error (e), namely $(\hat{\theta} - e, \hat{\theta} + e)$. In this case,

$$P(\hat{\theta} - e \leq \theta \leq \hat{\theta} + e) = 1 - \alpha$$

where $1 - \alpha$ is the probability that θ remains between the interval $(\hat{\theta} - e, \hat{\theta} + e)$. The interval $\hat{\theta} - e \leq \theta \leq \hat{\theta} + e$ is the $100(1 - \alpha)\%$ **confidence interval** for the population parameter θ . $1 - \alpha$ denotes the **confidence level for the interval estimation**.

1. The Confidence Interval for the Population Mean

a) If the population is **normally distributed** and its **variance is known**, from

$$P\left[-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right] = 1 - \alpha,$$

the confidence interval $100(1 - \alpha)\%$ for μ is

$$\left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right].$$

b) If the population is **normally distributed** and its **variance is unknown**, from

$$P\left[-t_{\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{\alpha/2, n-1}\right] = 1 - \alpha,$$

the confidence interval $100(1 - \alpha)\%$ for μ is

$$\left[\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}\right].$$

c) If the distribution of the population is **unknown** but its **variance is known**, when the size of the sample is **large** ($n \geq 30$), from

$$P\left[-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right] = 1 - \alpha,$$

the confidence interval $100(1 - \alpha)\%$ for μ is

$$\left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right].$$

d) If the **distribution and variance** of the population are **unknown**, when the size of the sample is **large** ($n \geq 30$), we can determine an approximate confidence interval from

$$P\left[-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq z_{\alpha/2}\right] = 1 - \alpha,$$

the confidence interval $100(1 - \alpha)\%$ for μ is

$$\left[\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}\right].$$

Example: It is known that the IQ scores of bachelor students in Turkey are distributed normally with a variance of 160. To estimate the mean of the distribution, 12 students are randomly selected, and their IQ scores are measured as follows:

103, 96, 112, 130, 111, 125, 89, 100, 104, 117, 99, 118

Construct a 95% confidence interval for the mean of all bachelor students in Turkey.

$$\bar{x} = \frac{\sum_{i=1}^{12} x_i}{12} = 108,67 \quad \sigma = \sqrt{160} \quad z_{\alpha/2} = z_{0,025} = 1.96$$

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad 108,67 - 1,96 \frac{\sqrt{160}}{\sqrt{12}} \leq \mu \leq 108,67 + 1,96 \frac{\sqrt{160}}{\sqrt{12}}$$

$$101,51 \leq \mu \leq 115,83$$

Thus, we can claim that the population mean μ is between the interval [101.51, 115.83] with a confidence of 95%.

Example: Solve the above problem in case that the population variance is unknown.

$$n=12, \quad t_{0,025,11} = 2.201 \quad s = \sqrt{\frac{\sum_{i=1}^{12} (x_i - \bar{x})^2}{11}} = 12.3 \quad \bar{x} = \frac{\sum_{i=1}^{12} x_i}{12} = 108,67$$

$$\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \quad 108,67 - 2,201 \frac{12,3}{\sqrt{12}} \leq \mu \leq 108,67 + 2,201 \frac{12,3}{\sqrt{12}}$$

$$100,85 \leq \mu \leq 116,49$$

Thus, we can claim that the population mean μ is between the interval [100.85, 116.49] with a confidence of 95%.

Example: To estimate the fuel consumption of a specific model of a car, 40 cars have been tested and it has been detected that the mean is 8.1 l/km and the standard deviation is 0.67 l/km. Construct a 95% confidence interval for the exact mean of all cars of the specific model.

$$\bar{x} = 8.1 \quad s = 0.67 \quad z_{\alpha/2} = z_{0,025} = 1.96 \quad n = 40$$

$$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}} \quad 8,1 - 1,96 \frac{0,67}{\sqrt{40}} \leq \mu \leq 8,1 + 1,96 \frac{0,67}{\sqrt{40}}$$

$$7,89 \leq \mu \leq 8,31$$

Thus, we can claim that the population mean μ is between the interval [7.89, 8.31] with a confidence of 95%.

2. The Confidence Interval for the Population Variance

If S^2 is the variance of a sample of size n that is selected from a population distributed normally, from

$$P \left[\chi^2_{1-\frac{\alpha}{2}, n-1} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\frac{\alpha}{2}, n-1} \right] = 1 - \alpha$$

the confidence interval $100(1-\alpha)\%$ for σ^2 is

$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right].$$

Example: The standard deviation of the fuel consumption of an engine that has been worked 16 times is 2.2 gallons. Construct a 99% confidence interval for the variance of the exact fuel consumption of the engine. (Assume that the population variance has a normal distribution.)

$$n = 16 \quad s = 2.2 \quad \chi^2_{0.005, 15} = 32.801 \quad \chi^2_{0.995, 15} = 4.601$$

$$\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \quad \frac{15(2.2)^2}{32.801} < \sigma^2 < \frac{15(2.2)^2}{4.601}$$

$$2.21 < \sigma^2 < 15.78$$

Thus, we can claim that the population variance σ^2 is between the interval [2.21, 15.78] with a confidence of 99%.

3. The Confidence Interval for the Population Proportion

If X is a random variable of a binomial distribution with parameters n and p , for the large values of n , from

$$P \left[-z_{\frac{\alpha}{2}} \leq \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \leq z_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

the confidence interval $100(1-\alpha)\%$ for p is

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\hat{p}\hat{q}/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\hat{p}\hat{q}/n} \right].$$

Example: 136 of 400 experimental people who were vaccinated for influenza have caught the disease. Construct a 95% confidence interval for the proportion of people who caught the disease among all vaccinated people.

$$n = 400 \quad \hat{p} = \frac{136}{400} = 0.34 \quad z_{0.025} = 1.96 \quad \hat{q} = 0.66$$

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} \quad 0.34 - 1.96 \sqrt{\frac{(0.34)(0.66)}{400}} \leq p \leq 0.34 + 1.96 \sqrt{\frac{(0.34)(0.66)}{400}}$$

$$0.294 \leq p \leq 0.386$$

Thus, we can claim that the population proportion p is between the interval $[0.294, 0.386]$ with a confidence of 95%.

In Case of Selection of a Sample without Replacement

When a sample is selected without replacement from a population of size N , and $n/N > 0.05$, the correction factor should be considered. In this case, since

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad \text{or} \quad \sigma_{\bar{x}} = \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \quad \sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

the confidence intervals for the means and proportion become as follows:

$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$	for normally distributed population with known variance
$\left[\bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$	for normally distributed population with unknown variance
$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$	for any population with known variance, $n > 30$
$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$	for any population with unknown variance, $n > 30$
$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}} \right]$	$n > 30$

4. The Confidence Interval for the Difference of Population Means

Consider that a sample of size n_1 whose members are

$$X_{11}, X_{12}, \dots, X_{1n_1}$$

is selected from a population with mean μ_1 and variance σ_1 , and similarly another sample of size n_2 whose members are

$$X_{21}, X_{22}, \dots, X_{2n_2}$$

is selected from a population with mean μ_2 and variance σ_2 . Then,

$$E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2,$$

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Theorem: If the two populations are distributed normally with the means μ_1 and μ_2 , the variances σ_1 and σ_2 , then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a standard normal distribution.

a) If $\bar{X}_1$ and $\bar{X}_2$ are two sample of sizes n_1 and n_2 which are respectively selected from two population distributed normally with variances σ_1 and σ_2 , then from

$$P \left[-z_{\frac{\alpha}{2}} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \leq z_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $\mu_1 - \mu_2$ is

$$\left[(\bar{X}_1 - \bar{X}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right].$$

Theorem: If the two populations (unknown distribution) have the means μ_1 and μ_2 , the variances σ_1 and σ_2 , then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$. Also, the population variances σ_1 and σ_2 are unknown, then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

has approximately the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$.

b) If $\bar{X}_1$ and $\bar{X}_2$ are two sample of sizes n_1 and n_2 which are respectively selected from two population (for any distribution) with variances σ_1^2 and σ_2^2 , then from

$$P \left[-z_{\frac{\alpha}{2}} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \leq z_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $\mu_1 - \mu_2$ is

$$\left[(\bar{X}_1 - \bar{X}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right].$$

c) Similarly, the population variances are unknown, from

$$P \left[-z_{\frac{\alpha}{2}} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \leq z_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $\mu_1 - \mu_2$ is

$$\left[(\bar{X}_1 - \bar{X}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right].$$

Theorem: If a random variable Z is normally distributed and independent χ_v^2 has a Chi-Squared distribution with the degree of freedom v , then the random variable

$$\frac{Z}{\sqrt{\frac{\chi_v^2}{v}}}$$

has a t distribution with degree of freedoms v .

We know from the theorem regarding variances that the random variables

$$\chi_1^2 = \frac{(n_1 - 1)S_1^2}{\sigma_1^2} \text{ and } \chi_2^2 = \frac{(n_2 - 1)S_2^2}{\sigma_2^2}$$

have Chi-Square distribution with the degrees of freedom $n_1 - 1$ and $n_2 - 1$ if populations are distributed normally with the means μ_1 and μ_2 and the variances σ_1^2 and σ_2^2 . If we take $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then the random variable

$$Y = \frac{(n_1 - 1)S_1^2}{\sigma^2} + \frac{(n_2 - 1)S_2^2}{\sigma^2} = \frac{(n_1 + n_2 - 2)S_k^2}{\sigma^2}$$

has also Chi-Square distribution with the degrees of freedom $n_1 + n_2 - 2$ where

$$S_k = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

is an unbiased estimator of the σ^2 .

Also, the random variable

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has the standard normal distribution. Now, by using the previous theorem, one can have the following theorem:

Theorem: If the two populations are distributed normally with the means μ_1 and μ_2 and their variances are equal, i.e. $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (but their values are unknown), then the random variable

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t distribution with degree of freedom $n_1 + n_2 - 2$. Here,

$$S_k = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

If the population variances are not equal, then the random variable

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

has a t distribution with degree of freedom v which can computed by rounding of

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}}.$$

d) If $\bar{X}_1$ and $\bar{X}_2$ are two sample of sizes n_1 and n_2 which are respectively selected from two population distributed normally with unknown variances $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then from

$$P \left[-t_{\frac{\alpha}{2}, n_1 + n_2 - 2} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \leq t_{\frac{\alpha}{2}, n_1 + n_2 - 2} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $\mu_1 - \mu_2$ is

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2}, n_1 + n_2 - 2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2}, n_1 + n_2 - 2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right].$$

e) Similarly, for the case unknown $\sigma_1^2 \neq \sigma_2^2$, from

$$P \left[-t_{\frac{\alpha}{2}, v} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \leq t_{\frac{\alpha}{2}, v} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $\mu_1 - \mu_2$ is

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right].$$

Example: To compare the nicotine content of two cigarette brands, 10 cigarettes are selected from the first brand and 8 cigarettes are selected from the second brand. The mean and standard deviation of the first sample are 3.1 mg and 0.5 mg, respectively, while the mean and standard deviation of the second sample are 2.7 mg and 0.7 mg, respectively. By assuming that both samples are selected from the populations distributed normally with equal variances, construct a 95% confidence interval for the difference between the two brand means.

$$n_1 = 10 \quad n_2 = 8 \quad s_1 = 0.5 \quad s_2 = 0.7$$

$$S_k = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{9(0.25) + 7(0.49)}{16}} = 0.596 \quad t_{0.025, 16} = 2.120$$

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2, n_1 + n_2 - 2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2, n_1 + n_2 - 2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$(3.1 - 2.7) - 2.12(0.596) \sqrt{\frac{1}{10} + \frac{1}{8}} \leq \mu_1 - \mu_2 \leq (3.1 - 2.7) + 2.12(0.596) \sqrt{\frac{1}{10} + \frac{1}{8}}$$

$$-0.20 \leq \mu_1 - \mu_2 \leq 1.0$$

Thus, we can claim that the difference between the two brands' means is between the interval $[-0.20, 1.0]$ with a confidence of 95%. Also, we cannot claim that there is a significant difference between means since the confidence interval contains zero.

Example: A study was conducted by the Department of Zoology at Virginia Tech to estimate the difference in the amounts of the chemical orthophosphorus measured at two different stations on the James River. Orthophosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average orthophosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligrams per liter. Find a 95% confidence interval for the difference in the mean of orthophosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

$$\bar{X}_1 = 3.84, \quad \bar{X}_2 = 1.49, \quad S_1 = 3.07, \quad S_2 = 0.80, \quad n_1 = 15, \quad n_2 = 12$$

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}} = \frac{\left(\frac{3.07^2}{15} + \frac{0.80^2}{12}\right)^2}{\frac{\left(\frac{3.07^2}{15}\right)^2}{15 - 1} + \frac{\left(\frac{0.80^2}{12}\right)^2}{12 - 1}} = 16.3 \approx 16, \quad \alpha = 0.05, \quad t_{\frac{\alpha}{2}, v} = t_{0.025, 16} = 2.120$$

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right]$$

$$= \left[(3.84 - 1.49) - 2.12 \sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}}, (3.84 - 1.49) + 2.12 \sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}} \right]$$

$$0.60 \leq \mu_1 - \mu_2 \leq 4.10$$

5. The Confidence Interval for the Difference of Population Proportions

If $\hat{p}_1 = \frac{X_1}{n_1}$ and $\hat{p}_2 = \frac{X_2}{n_2}$, then $E[\hat{p}_1 - \hat{p}_2] = p_1 - p_2$ and $\text{Var}[\hat{p}_1 - \hat{p}_2] = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$.

According to the Central Limit Theorem, the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - E[\hat{p}_1 - \hat{p}_2]}{\sqrt{\text{Var}[\hat{p}_1 - \hat{p}_2]}} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

has approximately distributed standard normally. For the large values of n_1 and n_2 , we can use $\hat{p}_1$ and $\hat{p}_2$ instead of p_1 and p_2 . Thus, the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

has also approximately distributed standard normally.

In this case, from

$$P \left[-z_{\frac{\alpha}{2}} \leq \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}} \leq z_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for $p_1 - p_2$ is

$$\left[(\hat{p}_1 - \hat{p}_2) - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}, (\hat{p}_1 - \hat{p}_2) + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \right].$$

Example: 132 of the 200 male voters and 90 of the 150 female voters support a specific candidate. Find a 99% confidence interval for the difference in the exact proportion of male and female voters that support that candidate.

$$\hat{p}_1 = \frac{132}{200} = 0.66 \quad \hat{q}_1 = \frac{68}{200} = 0.34 \quad \hat{p}_2 = \frac{90}{150} = 0.60 \quad \hat{q}_2 = 0.40 \quad z_{0.005} = 2.575$$

$$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \leq p_1 - p_2 \leq (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

$$(0.66 - 0.60) - 2.575 \sqrt{\frac{(0.66)(0.34)}{200} + \frac{(0.60)(0.40)}{150}} \leq p_1 - p_2 \leq$$

$$(0.66 - 0.60) + 2.575 \sqrt{\frac{(0.66)(0.34)}{200} + \frac{(0.60)(0.40)}{150}}$$

$$-0.074 \leq p_1 - p_2 \leq 0.194$$

The difference between the exact proportion of male and female voters that support that candidate is between the interval $[-0.074, 0.194]$ with a confidence of 99%.

6. The Confidence Interval for the Proportion of two variances

F Distribution

The probability density function of F distribution is given by

$$f(F; v_1, v_2) = \frac{\Gamma\left(\frac{v_1+v_2}{2}\right)}{\Gamma\left(\frac{v_1}{2}\right)\Gamma\left(\frac{v_2}{2}\right)} \left(\frac{v_1}{v_2}\right)^{\left(\frac{v_1}{2}\right)} F^{\left(\frac{v_1}{2}-1\right)} \left(1 + \frac{v_1}{v_2}F\right)^{-\left(\frac{v_1+v_2}{2}\right)}, F > 0$$

where positive integer v_1 and v_2 are called degree of freedom. The mean and variance of the random variable F is:

$$E(F) = \frac{v_2}{v_2 - 2}, \quad \text{Var}(F) = \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)},$$

F distribution denoted by F_{v_1, v_2} for the degree of freedom v_1 and v_2 .

Theorem: If S_1^2 and S_2^2 are the variances of the independent random samples of size n_1 and n_2 which are selected from two populations distributed normally with variances σ_1^2 and σ_2^2 , the random variable

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2}$$

has a F distribution with the degree of freedoms $n_1 - 1$ and $n_2 - 1$.

In this case, from

$$P \left[F_{1-\frac{\alpha}{2}, n_1-1, n_2-1} \leq \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \leq F_{\frac{\alpha}{2}, n_1-1, n_2-1} \right] = 1 - \alpha$$

the confidence interval $100(1 - \alpha)\%$ for σ_1^2 / σ_2^2 is

$$\left[\frac{1}{F_{\frac{\alpha}{2}, n_1-1, n_2-1}} S_1^2 / S_2^2, F_{\frac{\alpha}{2}, n_2-1, n_1-1} S_1^2 / S_2^2 \right].$$

Note that

$$F_{1-\frac{\alpha}{2}, n_1-1, n_2-1} = \frac{1}{F_{\frac{\alpha}{2}, n_2-1, n_1-1}}$$

Example: To compare the nicotine content of two cigarette brands, 10 cigarettes are selected from the first brand and 8 cigarettes are selected from the second brand. The mean and standard deviation of the first sample are 3.1 mg and 0.5 mg, respectively, while the mean and standard deviation of the second sample is 2.7 mg and 0.7 mg, respectively. Construct a 98% confidence interval for the proportion of the two brands' exact variances.

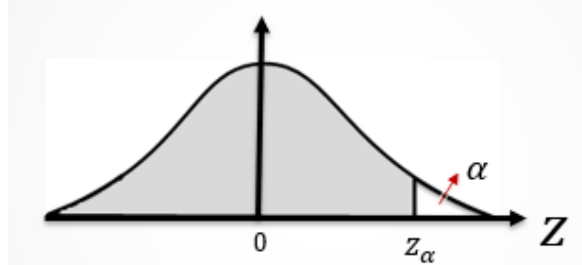
$$n_1 = 10 \quad n_2 = 8 \quad s_1 = 0.5 \quad s_2 = 0.7 \quad f_{0.01,9,7} = 6.72 \quad f_{0.01,7,9} = 5.61$$

$$\frac{0.25}{0.49} \frac{1}{6.72} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{0.25}{0.49} 5.61 \quad 0.076 < \frac{\sigma_1^2}{\sigma_2^2} < 2.862$$

The proportion of two brands' exact variances is between the interval [0.076, 2.862] with a confidence of 99%.

D. ONE-SIDED CONFIDENCE INTERVAL

It is possible to construct a one-sided confidence interval. Consider z distribution given below:



$$P(Z < Z_\alpha) = 1 - \alpha$$

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_\alpha\right) = 1 - \alpha$$

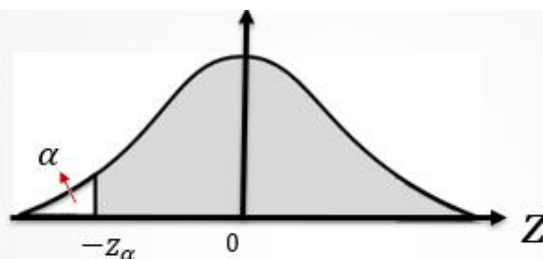
$$P\left(\mu > \bar{X} - Z_\alpha \sigma / \sqrt{n}\right) = 1 - \alpha \Rightarrow \left[\bar{X} - Z_\alpha \sigma / \sqrt{n}, \infty\right)$$

If the population has a normal distribution with a known variance, ...

Lower bound

One-sided (lower) confidence interval

For the left side,



$$P(Z > -Z_\alpha) = 1 - \alpha$$

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > -Z_\alpha\right) = 1 - \alpha$$

$$P\left(\mu < \bar{X} + Z_\alpha \sigma / \sqrt{n}\right) = 1 - \alpha \Rightarrow \left[-\infty, \bar{X} + Z_\alpha \sigma / \sqrt{n}\right)$$

If the population has a normal distribution with a known variance, ...

Upper bound

One-sided (upper) confidence interval

Similarly, one-sided confidence intervals can be derived for variances and proportions.

Example: In a psychological test experiment, the reaction times of 25 randomly selected subjects to a particular stimulus were measured. Past experiences show that the reaction time is normally distributed with a variance of 4 sec. The mean reaction time of 25 subjects is 6.2 sec. Compute the 95% upper bound for the mean reaction time.

$$\left[\bar{x} + z_{0.05} \frac{\sigma}{\sqrt{n}} = 6.2 + 1.645 \frac{\sqrt{4}}{\sqrt{25}} = 6.858 \text{ sec} \right]$$

$$\mu \in [-\infty, 6.858] \text{ for } \alpha = 0.05$$